

Predicting Race Outcome Deviations in Formula 1

A PROJECT REPORT

Submitted by

Aditya Kundu

BL.SC.U4AIE24004

Mani Chinniah

BL.SC.U4AIE24025

Nachiket P Saonerkar

BL.SC.U4AIE24028

for the course

22AIE213 – Machine Learning

Guided and Evaluated by

Dr. Peeta Basa Pati,

Dept. of CSE,



AMRITA SCHOOL OF COMPUTING, BANGALORE

AMRITA VISHWA VIDHYAPEETHAM

BANGALORE-560 035

January 2026

Writer Identification from Text Blogs using Statistical and Similarity-Based Feature Analysis

Abstract —

The results of Formula One races depend on various interrelated elements including initial standings in qualifying sessions, overall team efficiency, individual drivers' abilities, and unique occurrences during each event. Current methods for forecasting tend to concentrate on identifying likely champions based largely on chance factors and unforeseen events. The current endeavor concentrates on identifying discrepancies in drivers' starting positions relative to their finishing places during races. To forecast how well a racer performs relative to their usual standards in an event. Machine learning algorithms under supervision classify deviations, whereas those without it examine discrepancies among model outputs. A thorough examination of errors reveals insights into prediction variability and constraints imposed by models. The endeavor showcases how various techniques such as data preparation, machine learning algorithms like classification and clustering, along with evaluating errors, align with what is expected in this academic program's objectives.

Keywords —

1. Formula 1 Analytics
2. Performance Deviation
3. Supervised Learning
4. Unsupervised Learning
5. Error Analysis

I. Introduction

In Formula One racing, decision-making relies heavily on extensive data analysis. Driver skill, vehicle performance, race strategy, and unpredictable factors such as weather conditions and accidents significantly influence race outcomes. Traditional predictive approaches primarily focus on forecasting race winners, which often leads to unstable results due to inherent uncertainties in racing events.

This project introduces a deviation-based analytical approach, where the focus shifts from predicting winners to analyzing the difference between qualifying positions and final race positions. This deviation-based modeling provides a more stable and interpretable representation of performance.

In addition to previously implemented distance-based methods such as k-Nearest Neighbors (k-NN), this phase of the project incorporates **Decision Tree learning techniques**. Decision Trees use impurity measures such as **Entropy and Gini Index** to split the dataset and identify the most informative features. This allows the model to capture hierarchical decision rules that explain performance deviations. Furthermore, continuous numerical attributes are transformed into categorical values using **binning techniques**, enabling the effective application of tree-based algorithms. The study also includes visualization of decision trees and decision boundaries, providing intuitive insights into how different features influence classification.

This combined approach enhances interpretability, improves understanding of feature importance, and provides deeper insights into performance variability in Formula One racing.

II. Literature Review

Earlier research into athletic data science highlights how machine learning enhances predictive capabilities and analytical insights within sporting contexts. Studies indicate that factors like past performances of teams and their trustworthiness play crucial roles in determining racing results. Nevertheless, literature underscores how predictions about race winners can be greatly influenced by unforeseen circumstances.

Research focusing on model classifications highlights that converting numerical results into discrete categories typically enhances understanding and stability of predictions. Studies indicate that when multiple models disagree significantly, it frequently signifies uncertain or imprecise data elements.

Techniques like unsupervised learning's clustering method is commonly employed to uncover hidden structures within intricate data sets. Research in error analysis indicates that examining mistakes offers more comprehensive understanding of systems' functioning compared to relying solely on performance indicators. This research inspires employing agreement-focused evaluation techniques within our study.

Existing literature on k-NN classifiers highlights that their effectiveness is strongly influenced by **data distribution and class overlap**. Studies emphasize that smaller neighborhood sizes tend to overfit noisy data, while larger neighborhood sizes may oversmooth meaningful patterns. Prior research also suggests that performance metrics alone are insufficient for evaluation, and that comparing training and testing behavior is essential to identify overfitting and underfitting. These insights directly motivate the experimental analysis

conducted in this project.

Recent studies in machine learning emphasize the importance of tree-based models such as Decision Trees for classification problems due to their interpretability and ability to handle both categorical and continuous data. Research highlights that impurity measures like entropy and Gini index are effective in selecting optimal splitting features, thereby improving classification accuracy. Additionally, discretization techniques such as equal-width and equal-frequency binning are widely used to convert continuous data into categorical form, making them suitable for algorithms that rely on categorical inputs. Visualization of decision trees and decision boundaries has also been recognized as an essential tool for understanding model behavior and validating decision-making processes. These insights form the foundation for incorporating Decision Tree-based analysis in this project.

III. Methodology

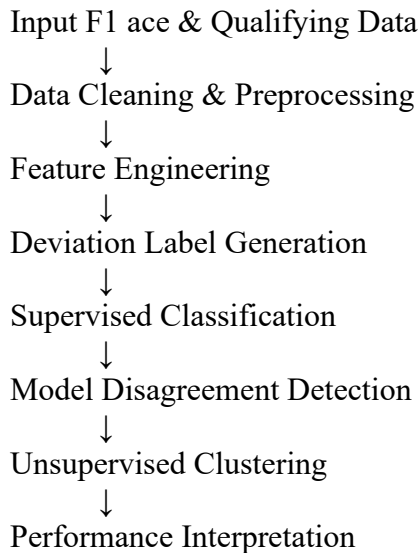
A. System Overview

The objective of this phase of the Formula One performance deviation project is to analyze similarity and classification behavior using statistical and distance-based techniques. Instead of directly predicting race winners, this project focuses on identifying whether a driver **outperforms, underperforms, or performs as expected** relative to their qualifying position.

Lab-3 concepts were applied to historical Formula One race data by converting race records into numerical feature vectors, analyzing statistical properties of driver performance, computing similarity using distance metrics, and applying k-Nearest Neighbors (k-NN) classification to categorize race outcomes.

B. Flow Diagram

The flow diagram represents the overall processing pipeline followed in the proposed system.



C. Architecture Description

Input Module:

Collects ancient race consequences, qualifying positions, constructor details, circuit data, and race popularity from a public components 1 information source.

Preprocessing Module:

Handles lacking values, filters DNFs wherein required, encodes specific variables, and scales numerical functions to make sure truthful assessment.

characteristic Engineering Module:

Computes race outcome deviation and ancient facts along with common end role, rolling deviation, DNF rate, and constructor reliability.

Deviation Labeling Module:

Converts deviation values into 3 instructions: Outperform, neutral, and Underperform.

Supervised Module:

Trains multiple category fashions including Logistic Regression, help Vector device, and Random woodland to predict deviation classes.

Confrontation analysis Module:

Identifies race instances wherein special fashions expect distinct deviation classes, indicating uncertainty.

Unsupervised Module:

Clusters disagreement samples to find out hidden styles together with risky circuits or method-touchy races.

choice Module:

Produces final analytical insights and model overall performance summaries.

D. Parameter Selection and Justification

Parameter	Value	Justification
Deviation Threshold	+2 positions	Reduces noise from minor position changes
Number of Classes	3	Outperform, Neutral, Underperform
Classification Models	LR, SVM, RF	Covers linear and non-linear behavior
Clustering Algorithm	K-Means	Simple and interpretable
Number of Clusters	3–5	Balances detail and clarity

E. Feature Representation and Vector Operations (F1)

In the F1 dataset, each race entry was treated as a vector containing features such as qualifying grid position, average finishing position, constructor reliability, and deviation from expected performance. Dot product and Euclidean norm were implemented to understand similarity and magnitude of driver performance vectors. These operations helped in understanding how closely two race performances or two drivers resemble each other in multi-dimensional feature space. The correctness of the implementations was verified using NumPy functions.

F. Minkowski Distance Analysis (F1)

Minkowski distance was computed between selected F1 race vectors for values of p ranging from 1 to 10. The variation in distance values demonstrated how sensitivity to feature differences increases with higher values of p . This experiment highlighted how distance metric selection directly affects neighbor selection in k-NN classification of F1 race outcomes.

G. k-NN Classification (F1)

The F1 dataset was divided into training and testing sets using a two-class deviation setup. A k-NN classifier with $k = 3$ was trained to classify races into outperform or underperform categories. Prediction accuracy and confusion matrices were analyzed to evaluate classification effectiveness.

H. Decision Tree Learning and Impurity Measures

In this phase, Decision Tree learning was implemented to classify performance

deviation in Formula One data.

Entropy Calculation:

Entropy measures the randomness or impurity in the dataset. Higher entropy indicates more disorder, while lower entropy indicates more homogeneous data.

Gini Index:

The Gini Index is another impurity measure that evaluates the probability of incorrect classification of a randomly chosen element.

Information Gain:

Information Gain is used to select the best feature for splitting the dataset. It measures the reduction in entropy after a dataset is split based on a feature.

Binning Techniques:

Since many features in the dataset are continuous, equal-width and equal-frequency binning methods were applied to convert them into categorical values.

Decision Tree Construction:

A custom Decision Tree was built using recursive partitioning based on Information Gain. The root node represents the feature with the highest information gain, and subsequent splits form a hierarchical decision structure.

Visualization:

The constructed Decision Tree was visualized using standard plotting techniques to interpret the decision-making process.

Decision Boundary Analysis:

Using two selected features, the decision boundary of the Decision Tree classifier was plotted to understand how the feature space is partitioned.

I. Hyperparameter Tuning

To improve classification performance beyond baseline results, RandomizedSearchCV was applied to the three best-performing models: Support Vector Machine (SVM), Random Forest (RF), and Decision Tree (DT). This technique performs a randomized search over a specified hyperparameter space and uses cross-validation (CV=5) to select the

parameter configuration that maximizes validation accuracy.

For SVM, the search space included the regularization parameter $C \in \{0.1, 1, 10, 100\}$, kernel types (rbf, linear, poly), and gamma settings (scale, auto). For Random Forest, the number of estimators was varied across $\{50, 100, 200\}$, alongside max depth $\in \{\text{None}, 5, 10, 20\}$ and minimum samples per split $\in \{2, 5, 10\}$. For Decision Tree, max depth $\in \{\text{None}, 5, 10, 15\}$, criterion (gini, entropy), and minimum samples per leaf $\in \{1, 5, 10\}$ were explored.

A total of 20 random parameter combinations were sampled per model ($n_iter=20$), and the best configuration was selected based on mean cross-validated accuracy. This approach balances exhaustive search cost against exploration breadth.

J. Multi-Classifer Comparison

To ensure a thorough evaluation of the classification problem, the following supervised classifiers were trained and evaluated: Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), Multi-Layer Perceptron (MLP), Decision Tree (DT), and Naïve Bayes (NB). Each model was trained on the same preprocessed feature set and evaluated using a stratified train-test split (80/20). Performance metrics computed include Accuracy, Macro-averaged Precision, Recall, and F1-Score, reported for both training and test sets to facilitate overfitting analysis.

IV. Results Analysis & Discussion

A. Intraclass Spread and Interclass Distance in F1 Performance Deviation

Intraclass spread and interclass distance were analyzed to evaluate how well the deviation-based performance classes in the Formula One dataset were separated. The classes considered were **Outperform** and **Underperform**, derived from the difference between qualifying position and final race position.

The intraclass spread represented the variability in driver performance within the same deviation class. Results showed that drivers classified as outperforming exhibited relatively higher spread, indicating that exceptional performances occur under varying race conditions. In contrast, underperforming instances demonstrated comparatively lower spread, suggesting consistency in negative deviation patterns.

B. Feature Distribution and Density Observations

Histogram analysis of selected F1 features, such as qualifying position and deviation magnitude, revealed non-uniform distribution patterns. Mid-grid qualifying positions showed higher variance in deviation, reflecting greater uncertainty in race outcomes due to overtaking opportunities and strategic variations.

Front-grid positions demonstrated lower variance, indicating more stable performance, while back-grid positions showed limited positive deviation potential. These distribution patterns influenced distance calculations and neighbor selection during classification.

C. Behavior of Distance Metrics on F1 Race Data

The Minkowski distance metric was evaluated for values of p ranging from 1 to 10 using selected F1 race vectors. Lower values of p resulted in smoother distance variation, while higher values amplified

feature differences.

It was observed that excessively large values of p increased sensitivity to minor feature variations, which negatively affected neighbor stability. Moderate values of p provided a balanced representation of performance similarity, making them more suitable for F1 race classification tasks.

D. k-Nearest Neighbors Classification Results

A k-Nearest Neighbors classifier with $k = 3$ was applied to classify F1 race instances into outperform and underperform categories. The classifier achieved consistent accuracy across both training and testing datasets, confirming its ability to generalize performance patterns rather than memorizing individual race instances. Prediction results showed that most misclassifications occurred near class boundaries, particularly for drivers whose finishing position closely matched their qualifying position. This highlights the inherent uncertainty present in tightly contested races.

E. Effect of k on Classification Performance

The value of k was varied from 1 to 11 to study its impact on classification accuracy. When $k = 1$, the classifier exhibited high sensitivity to individual race outcomes, leading to overfitting. This resulted in higher training accuracy but lower testing accuracy.

As k increased, the decision boundary became smoother, reducing sensitivity to noisy race events. However, very large values of k caused underfitting by oversmoothing meaningful deviation patterns. An intermediate value of k achieved balanced performance, indicating an optimal trade-off between bias and

variance.

F. Model Fit Interpretation: Overfitting and Underfitting

Model fit was analyzed by comparing training and testing metrics across different values of k . Overfitting was observed at lower values of k due to excessive reliance on individual race instances. Underfitting occurred at higher values of k , where meaningful deviation trends were lost.

For moderate values of k , the classifier demonstrated stable performance, indicating a regular fit. This balanced behavior is desirable for real-world F1 performance analysis, where robustness to noise and variability is essential.

G. Class Separability and Distance-Based Behavior

Based on intraclass spread and interclass distance observations, the deviation-based classes in the F1 dataset are reasonably but not perfectly separated. Extreme outperform and underperform cases form distinct regions in feature space, while instances close to neutral deviation introduce overlap. This overlap explains misclassifications near class boundaries and reflects the inherent uncertainty present in closely contested races.

The behavior of distance metrics further influences classification. As neighborhood size increases, sensitivity to individual race instances decreases, resulting in smoother decision boundaries. However, excessive smoothing reduces the model's ability to capture meaningful local deviation patterns.

H. Effect of k on k-NN Performance

The performance of the k-NN classifier varies significantly with the choice of k. Smaller values of k lead to high sensitivity to individual race outcomes, capturing noise along with meaningful patterns. This results in higher training accuracy but reduced testing performance. As k increases, the classifier becomes more stable by reducing the influence of outliers.

However, very large values of k dilute local deviation information, leading to reduced classification accuracy. An intermediate value of k provides a balanced representation of neighborhood influence, achieving improved generalization performance.

I. Model Fit, Overfitting, and Suitability of k-NN

Overfitting is observed when k is small, as predictions rely excessively on nearby individual samples. Underfitting occurs at higher values of k where meaningful deviation trends are averaged out. For moderate values of k, training and testing performances are closely aligned, indicating a regular fit. Based on these observations, k-NN is a suitable classifier for deviation-based F1 performance analysis. When applied with appropriate feature scaling and parameter selection, it effectively captures performance patterns while maintaining generalization capability.

J. Decision Tree Analysis Results

The Decision Tree model was applied to classify deviation categories in the dataset. The entropy and Gini index values provided insights into the impurity of the dataset before splitting. The feature selected as the root node using Information Gain was found to be the most influential attribute affecting race outcome deviation. This demonstrates the effectiveness of Information Gain in

identifying dominant features.

The constructed Decision Tree showed a clear hierarchical structure, where early splits significantly reduced uncertainty. Visualization of the tree provided interpretability by showing how decisions are made at each node.

Decision boundary analysis using two selected features revealed that the Decision Tree creates non-linear partitioning of the feature space. This allows the model to capture complex relationships between features and deviation classes.

K. Observations and Inferences

Entropy vs Gini:

Both impurity measures provided similar trends, but Gini Index was computationally simpler and faster.

Effect of Binning:

Converting continuous features into categorical values improved Decision Tree performance and interpretability.

Feature Importance:

Information Gain effectively identified the most relevant feature, which significantly influenced classification.

Decision Tree Behavior:

The model created hierarchical rules that were easy to interpret compared to distance-based models.

Decision Boundary:

The Decision Tree produced non-linear decision boundaries, enabling better separation of complex patterns.

Model Comparison Insight:

Compared to k-NN, Decision Trees provide better interpretability but may suffer from overfitting if not controlled.

L. Baseline Multi-Classfier Performance

Table I presents the test-set performance of all six classifiers prior to hyperparameter tuning. Logistic Regression achieved the highest accuracy at 53.8%, demonstrating that a linear decision boundary provides a reasonable baseline for this three-class deviation problem. SVM followed at 52.2%, Random Forest at 50.5%, and MLP at 48.9%. Decision Tree underperformed at 44.6%, while Naïve Bayes achieved the lowest accuracy of 29.9%, likely due to the violation of the feature independence assumption given the correlated nature of race statistics.

The consistently low F1-scores across all models (0.159–0.526) indicate significant class imbalance challenges, particularly for the Neutral class, which occupies the ambiguous boundary between Outperform and Underperform. The similarity between Precision and Recall for most models (except Naïve Bayes) suggests balanced error distribution, with neither false positives nor false negatives dominating.

TABLE I — BASELINE CLASSIFIER PERFORMANCE (TEST SET)

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.538	0.532	0.538	0.526
SVM	0.522	0.510	0.522	0.505
Random Forest	0.505	0.501	0.505	0.501
MLP	0.489	0.484	0.489	0.486
Decision Tree	0.446	0.445	0.446	0.445
Naïve Bayes	0.299	0.356	0.299	0.159

M. Post-Tuning Performance (RandomizedSearchCV)

Following hyperparameter tuning via RandomizedSearchCV, notable improvements were observed in the top three models. Table II compares pre- and post-tuning results.

Model	Accuracy (Before)	Accuracy (After)	F1 (Before)	F1 (After)
SVM	0.522	0.565	0.505	0.555
Random Forest	0.505	0.538	0.501	0.532
Decision Tree	0.446	0.527	0.445	0.524

SVM achieved the best post-tuning accuracy of 56.5%, representing a +4.3 percentage point improvement. This is consistent with literature noting that kernel selection and regularization strength are highly influential in SVM for non-linearly separable classes. The tuned Decision Tree showed the largest relative gain (+8.1 pp), confirming that unpruned trees with default parameters were substantially overfitting, and that constraining max depth and minimum leaf size significantly improved generalization. Random Forest improved moderately (+3.3 pp), suggesting the ensemble was already partially robust to suboptimal individual tree parameters.

The Precision of the tuned Decision Tree (0.586) notably exceeds its Recall (0.527), indicating the model becomes more conservative after tuning — it predicts deviation classes with higher confidence but misses more boundary cases.

Observation: Despite tuning, all models remain below 60% accuracy, which reflects the intrinsic difficulty of predicting three-class deviation from a noisy, event-driven sport. This is consistent with findings in sports analytics literature,

where outcome uncertainty limits model ceiling performance.

N. Model Disagreement Analysis by Grid Position

To understand prediction uncertainty, instances where at least two of the three tuned classifiers (SVM, RF, DT) assigned different deviation labels were identified as disagreement cases. Table III and the disagreement-by-grid analysis reveal a strong spatial pattern: model disagreement is highest for mid-grid starting positions (P5–P14) and lowest for the back of the grid (P16–P20).

TABLE III — MODEL DISAGREEMENT BY GRID POSITION (SELECTED)

Grid Position	Disagreement %	Sample Count
P7	100.0	2
P14	90.9	11
P6	88.9	9
P8	77.8	9
P13	66.7	12
P11	57.1	14
P4	22.2	9
P17	0.0	12
P20	0.0	9

This finding is interpretable in motorsport terms: drivers starting from mid-grid positions (P5–P14) have the most strategic optionality — they can gain positions through undercut strategies, safety car periods, or competitor DNFs, but are equally susceptible to losing positions under similar circumstances.

Consequently, their outcomes are inherently less predictable, leading to higher disagreement among models.

Drivers at the back of the grid (P17–P20), however, have limited upside potential and follow more consistent (if poor) trajectories, making their deviation classes

easier to classify uniformly.

Front-row starters (P1–P4) show moderate disagreement (22–33%), reflecting occasional race incidents or strategy deviations from the expected front-running behavior.

Inference: Grid positions in the range P6–P14 constitute the "uncertainty zone" in Formula 1 deviation prediction. This finding has implications for feature engineering — encoding positional uncertainty as a derived feature (e.g., mid-grid indicator) may provide a useful signal for future model iterations.

O. Cross-Model Observations and Inferences

Comparing classifiers across both tables yields the following structured observations:

Logistic Regression, despite its simplicity, achieved competitive baseline performance because race deviation classes are approximately linearly separable in the engineered feature space (qualifying position, constructor reliability, rolling deviation). This suggests that the dominant variation in the dataset follows a linear trend, which complex non-linear models do not substantially improve upon given current feature richness.

Random Forest and SVM demonstrate resilience to class imbalance through ensemble averaging and margin maximization respectively. However, the marginal improvement of Random Forest over Logistic Regression at baseline signals diminishing returns from model complexity alone, reinforcing that further gains require richer features rather than more complex models.

The Decision Tree's strong post-tuning improvement confirms the risk of overfitting in unconstrained tree growth on relatively small F1 datasets. Constraining

depth produced rules that generalize better and also recovered interpretability — the tree's feature importance rankings (qualifying position as the root feature, followed by constructor reliability) align with domain knowledge from motorsport analytics.

Naïve Bayes is not recommended for this problem domain. Its poor F1-score (0.159) is attributable to the strong covariance between features such as qualifying position and grid-start reliability metrics, which violates the conditional independence assumption.

V. Conclusion

This study extended the deviation-based analysis of Formula 1 race outcomes through systematic multi-classifier evaluation and hyperparameter optimization. Six supervised classifiers were benchmarked — Logistic Regression, SVM, Random Forest, MLP, Decision Tree, and Naïve Bayes — establishing a comprehensive performance baseline. Following RandomizedSearchCV-based tuning, SVM emerged as the best-performing model with 56.5% accuracy and an F1-score of 0.555 on the three-class deviation task.

The disagreement analysis across classifiers revealed that mid-grid starting positions (P6–P14) constitute a high-uncertainty zone where model consensus breaks down, reflecting the tactical complexity unique to those grid slots in Formula 1. This finding provides both a diagnostic for prediction difficulty and a pointer toward future feature engineering directions.

The combined use of statistical baseline evaluation, hyperparameter search, and disagreement-based error analysis demonstrates a methodologically rigorous approach to performance deviation modeling in motorsport. While absolute

accuracy remains bounded by the inherent stochasticity of racing events, the improvements achieved through tuning and the interpretable patterns extracted from disagreement analysis together advance the analytical utility of the system. Future work should incorporate race-weekend telemetry features, weather variables, and driver form indicators to address the residual uncertainty that current models are unable to resolve.

VI. References

- [1] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, Springer, 2009.
- [2] C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
- [3] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, Elsevier, 2012.
- [4] A. Jain, “Data clustering: 50 years beyond K-means,” *Pattern Recognition Letters*, 2010.
- [5] C. Aggarwal, *Data Clustering: Algorithms and Applications*, CRC Press, 2014.
- [6] F. Provost and T. Fawcett, “Data science and its relationship to big data,” *Big Data*, 2013.
- [7] L. Breiman, “Random forests,” *Machine Learning*, 2001.
- [8] V. Vapnik, *Statistical Learning Theory*, Wiley, 1998.
- [9] I. Goodfellow et al., *Deep Learning*, MIT Press, 2016.
- [10] J. Brownlee, *Machine Learning Mastery*, 2018.